Lunar Relics 3D — Project Description

NASA Space Apps Challenge 2026 · Cairo Challenge: Abandoned but not Forgotten: Storytelling about NASA's Discarded Equipment on the Moon and Mars **Project name:** Lunar Relics 3D

	Link
Live project (open directly, no login)	https://lunar-relics-3d.vercel.app/
30-second demo video	https://lunar-relics-3d.vercel.app/demo/lunar-relics-3d-demo-30s.mp4
Public source code	https://github.com/yousofahmad/lunar-relics-3d
This document	https://lunar-relics-3d.vercel.app/docs/LunarRelics3D_Project_Description.pdf

1. High-Level Summary

Lunar Relics 3D is a bilingual Arabic–English interactive archive of 16 NASA mission relics across the Moon, Mars, and deep space. It turns dispersed mission records into first-person spacecraft stories, navigable 3D scenes, timelines, hardware hotspots, recorded narration, quick quizzes, badges, and a four-mode Explorer Challenge. Eleven missions use NASA-published 3D meshes; the other five use SI-generated models built to NASA's published specifications and clearly labeled "not from NASA", with source links kept visible. The site is responsive, shareable by mission link, and designed for students, families, and casual space learners. It addresses the challenge by making discarded hardware understandable as evidence: what it was built to do, where it went, what it measured, and what remains after the mission ends.

2. Project Details

What it does

Lunar Relics 3D is a browser-based, bilingual learning experience about equipment that remained after missions to the Moon, Mars, and beyond. The archive contains 16 structured records. Each record connects a relic to its mission dates, destination, status, source links, first-person English and Arabic story, timeline, hardware explanations, questions, quiz, badge, and recorded narration. The voice is a storytelling device: the spacecraft speaks in first person, while the interface makes clear which facts and sources support the story.

Visitors begin at a 3D globe-style map with mission pins and can filter the archive by world or 3D availability. Selecting a mission opens a chaptered story beside its 3D scene. Hardware hotspots identify parts and draw the learner toward an explanation of what each part did. The mission timeline then connects hardware to journey and scientific purpose. A quick quiz awards one locally saved badge per mission. The Explorer Challenge adds four playable modes: evidence quiz, mission timeline ordering, launch-era ordering, and "name the machine" model identification.

Guided, accessible exploration

Audio is optional. Every mission has English and Arabic recorded narration for the main story, an introduction, and the hardware points represented in the narration manifest. "Full journey" starts with the introduction, moves through the story and hardware chapters, follows the page while the voice plays, reaches the timeline, and ends at the quiz. Learners can stop, scroll manually, choose a language, or use the text-only path. Mission links encode the mission, world, and language, so a learner can bookmark or share a specific story.

Accuracy and provenance

The project is intentionally honest about representation. Eleven missions use unmodified NASA-published GLB meshes from NASA 3D Resources and are labeled as NASA models. The five missions that have no public NASA mesh (Apollo 15 Lunar Roving Vehicle, Apollo 12 ALSEP, Surveyor 3, Sojourner, Phoenix) use 3D models generated with Super Intelligence from written descriptions of NASA's published specifications and photographs; every one of them carries the on-screen label "SI-generated model built to published specs · not from NASA", and the repository's asset-source file lists them separately. The Spirit entry uses the NASA Opportunity mesh as a representative model of the identical rover design; Pioneer 10 is used as the available model for a record that discusses Pioneer 10 and 11. These limitations are shown rather than hidden. The Moon and Mars globe textures are stored locally, and the Mars height treatment is explicitly visual rather than a scientific elevation model.

Technology and intended impact

The front end uses React 18 and Vite. Three.js, @react-three/fiber, and drei provide the 3D scenes; Tailwind/PostCSS support styling; lucide-react supplies interface icons; canvas-confetti supplies reduced-motion-aware feedback. The Web Audio API handles the

optional soundscape and narration routing. GitHub hosts the public source and Vercel hosts the live site. FFmpeg supports audio/video preparation, while Python and Playwright were used for testing and screenshot capture. The repository includes a production build and Node tests for selected experience and asset-integrity behaviors, without claiming exhaustive device certification.

Creatively, the project treats discarded equipment as a collection of voices and evidence rather than a static list of names. It is built for school-age learners, families, teachers, and casual space enthusiasts, especially users who prefer Arabic or need a low-friction phone experience. The team considered RTL layout, bilingual labels and subtitles, keyboard and skip-to-content access, reduced-motion behavior, responsive breakpoints, lazy model loading, capped device pixel ratio, local textures, and a local JSON fallback when an optional API is unavailable. The intended impact is a durable, source-linked path from “what was left behind?” to “what did this machine enable us to learn?”

3. How to Use It (for judges)

1. Open https://lunar-relics-3d.vercel.app/ in any modern browser (desktop or phone). No account is needed.
2. Choose English or عربي on the opening screen, then press **Start with the Eagle** or pick the Moon, Mars, or Deep Space.
3. Click any mission pin (or open **Missions** in the top bar) to open its story. Scroll: each wheel step moves exactly one chapter — Start, Story, hardware points, Journey, Quick Quiz.
4. Press **Begin the story** (or **Auto-pilot** in the top bar) for the fully narrated journey; audio is optional and subtitles are always shown.
5. On a hardware chapter, the red pin and leader line point to the part on the 3D model; press **Listen** for its narration.
6. Open **Explorer Challenge** in the top bar to play the four game modes; finish a mission quiz to earn its badge.

4. Working Features, Limitations, and Future Plans

Working now: 16 bilingual (Arabic/English) mission records across the Moon, Mars and deep space; interactive 3D map and mission scenes; first-person stories with recorded Arabic and English narration for the intro, story and every hardware point; chapter-by-chapter scrolling; hardware hotspots with leader lines; mission timelines; quick quizzes and 16 badges; Explorer Challenge with four modes (evidence quiz, timeline ordering, launch-era ordering, name-the-machine); shareable mission links; responsive desktop and mobile layout; RTL Arabic layout; reduced-motion support.

Limitations: five missions use SI-generated models rather than NASA meshes (clearly labelled); Spirit and Viking 2 reuse a sister-craft NASA mesh; map positions and lighting are for exploration, not precision measurement; the Mars relief is visual, not a scientific elevation model; progress and badges are stored locally in the browser; heavy 3D scenes can be slow on very old phones.

Future plans: add more relics (Luna, Chang’e, Apollo 12–17 hardware, MSL Curiosity cruise stage); a classroom mode with printable mission cards; teacher-facing source notes; expert review of the Arabic science vocabulary; offline/PWA support; replacing generated models with official meshes as NASA publishes them.

5. NASA Data

How NASA data is used in the project

- **NASA 3D Resources (GLB meshes):** 11 of the 16 missions are shown with unmodified NASA-published 3D models (Apollo Lunar Module, Opportunity, Viking Lander, InSight, Ingenuity, Dawn, NEAR Shoemaker, Pioneer 10, Kepler). Spirit reuses the Opportunity mesh (identical rover design); Viking 1 and 2 share the Viking Lander mesh.
- **NSSDCA spacecraft catalog records (NASA Space Science Data Coordinated Archive):** the source of each record’s NSSDCA ID, launch date, landing/arrival data and mission facts used in the story, timeline, quiz and Explorer Challenge.
- **NASA Science, NASA.gov and JPL mission pages and news releases:** facts for the first-person stories, hardware hotspot descriptions, end-of-mission dates and the “what remains” sections. Every record keeps its source links in the data file.
- **NASA SVS CGI Moon Kit (LRO LROC colour + LOLA elevation):** used for the Moon globe texture and relief on the mission map.
- **NASA photographs and published specifications:** used as the written reference for the five SI-generated models (Apollo 15 LRV, ALSEP, Surveyor 3, Sojourner, Phoenix), which are labelled “not from NASA”.
- **Mars globe:** USGS Astrogeology Viking colorized global mosaic (built from NASA Viking Orbiter imagery), listed under other data.

The following are the deduplicated NASA resources used or cited by the 16 mission records, model provenance notes, and local texture notes. The links are grouped by source family for easier form entry.

NASA resource and asset provenance

Link Text	Link Url
assets.science.nasa.gov/content/dam/science/cds/3d/resources/model/apollo-lunar-module/Apollo%20Lunar%20Module.glb	https://assets.science.nasa.g module/Apollo%20Lunar%20
assets.science.nasa.gov/content/dam/science/psd/mars/resources/gltf_files/24880_InSight_deployed.glb	https://assets.science.nasa.g
github.com/nasa/NASA-3D-Resources	https://github.com/nasa/NASA
github.com/nasa/NASA-3D-Resources/tree/master/3D%20Models/Dawn	https://github.com/nasa/NASA
github.com/nasa/NASA-3D-Resources/tree/master/3D%20Models/InSight%20Cruise%20Lander	https://github.com/nasa/NASA
github.com/nasa/NASA-3D-Resources/tree/master/3D%20Models/Ingenuity%20Mars%20Helicopter	https://github.com/nasa/NASA
github.com/nasa/NASA-3D-Resources/tree/master/3D%20Models/Kepler%20%28B%29	https://github.com/nasa/NASA
github.com/nasa/NASA-3D-Resources/tree/master/3D%20Models/Mars%20Exploration%20Rover%20-%20Opportunity%20%28MER-B%29	https://github.com/nasa/NASA %20Opportunity%20%28MEF
github.com/nasa/NASA-3D-Resources/tree/master/3D%20Models/NEAR%20Shoemaker	https://github.com/nasa/NASA
github.com/nasa/NASA-3D-Resources/tree/master/3D%20Models/Viking%20Lander	https://github.com/nasa/NASA
NASA 3D Resource — 3D Resources	https://science.nasa.gov/3d-re
NASA 3D Resource — Apollo Lunar Module	https://science.nasa.gov/3d-re
NASA 3D Resource — Dawn Spacecraft	https://science.nasa.gov/3d-re
NASA 3D Resource — Ingenuity Mars Helicopter	https://science.nasa.gov/3d-re
NASA 3D Resource — Insight Cruise Lander	https://science.nasa.gov/3d-re
NASA 3D Resource — Kepler B	https://science.nasa.gov/3d-re
NASA 3D Resource — Mars Exploration Rover Opportunity Mer B	https://science.nasa.gov/3d-re
NASA 3D Resource — Near Shoemaker	https://science.nasa.gov/3d-re
NASA 3D Resource — Pioneer	https://science.nasa.gov/3d-re
NASA 3D Resource — Viking Lander	https://science.nasa.gov/3d-re
NASA Science resource — Insight Lander 3D Model	https://science.nasa.gov/reso
svs.gsfc.nasa.gov/4720/	https://svs.gsfc.nasa.gov/472
svs.gsfc.nasa.gov/vis/a000000/a004700/a004720/ldem_16.tif	https://svs.gsfc.nasa.gov/vis/a
svs.gsfc.nasa.gov/vis/a000000/a004700/a004720/lroc_color_16bit_srgb_4k.tif	https://svs.gsfc.nasa.gov/vis/a
www.nasa.gov/nasa-brand-center/images-and-media/	https://www.nasa.gov/nasa-br

NASA mission, archive, science, and JPL sources from the mission records

Link Text	Link Url
mars.nasa.gov/technology/helicopter/	https://mars.nasa.gov/technology/helicopter/
NSSDCA spacecraft catalog record — 1967-035A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1967-035A
NSSDCA spacecraft catalog record — 1969-059C	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1969-059C
NSSDCA spacecraft catalog record — 1969-099C	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1969-099C
NSSDCA spacecraft catalog record — 1971-008C	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1971-008C
NSSDCA spacecraft catalog record — 1971-063C	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1971-063C
NSSDCA spacecraft catalog record — 1972-012A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1972-012A
NSSDCA spacecraft catalog record — 1972-031C	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1972-031C
NSSDCA spacecraft catalog record — 1972-096C	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1972-096C
NSSDCA spacecraft catalog record — 1973-019A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1973-019A
NSSDCA spacecraft catalog record — 1975-075C	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1975-075C
NSSDCA spacecraft catalog record — 1975-083C	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1975-083C
NSSDCA spacecraft catalog record — 1996-008A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1996-008A
NSSDCA spacecraft catalog record — 1996-068A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1996-068A
NSSDCA spacecraft catalog record — 2003-032A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2003-032A
NSSDCA spacecraft catalog record — 2007-034A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2007-034A
NSSDCA spacecraft catalog record — 2007-043A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2007-043A
NSSDCA spacecraft catalog record — 2009-011A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2009-011A
NSSDCA spacecraft catalog record — 2018-042A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2018-042A
NSSDCA spacecraft catalog record — 2020-052A	https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2020-052A
nssdc.gsfc.nasa.gov/nmc/spacecraftDisplay.do?id=1971-063C	https://nssdc.gsfc.nasa.gov/nmc/spacecraftDisplay.do?id=1971-063C
NSSDCA planetary archive — chronology_mars.html	https://nssdc.gsfc.nasa.gov/planetary/chronology_mars.html
NSSDCA planetary archive — apollo11info.html	https://nssdc.gsfc.nasa.gov/planetary/lunar/apollo11info.html
NSSDCA planetary archive — apollo_lrv.html	https://nssdc.gsfc.nasa.gov/planetary/lunar/apollo_lrv.html
NSSDCA planetary archive — apolloloc.html	https://nssdc.gsfc.nasa.gov/planetary/lunar/apolloloc.html
NSSDCA planetary archive — lunar_sites.html	https://nssdc.gsfc.nasa.gov/planetary/lunar/lunar_sites.html
NSSDCA planetary archive — mera_images.html	https://nssdc.gsfc.nasa.gov/planetary/mars/mars_exploration_rovers/mera_images.
NSSDCA planetary archive — mars_2003_05.html	https://nssdc.gsfc.nasa.gov/planetary/mars_2003_05.html
NSSDCA planetary archive — marsland.html	https://nssdc.gsfc.nasa.gov/planetary/marsland.html
NSSDCA planetary archive — near.html	https://nssdc.gsfc.nasa.gov/planetary/near.html
NSSDCA planetary archive — viking.html	https://nssdc.gsfc.nasa.gov/planetary/viking.html

Link Text	Link Url
science.nasa.gov/astrobiology/missions/viking-project/	https://science.nasa.gov/astrobiology/missions/viking-project/
NASA Earth Observatory — Apollo 11 Landing Site 39408	https://science.nasa.gov/earth/earth-observatory/apollo-11-landing-site-39408/
NASA Science mission page — Dawn	https://science.nasa.gov/mission/dawn/
NASA Science mission page — Faq	https://science.nasa.gov/mission/dawn/faq/
NASA Science mission page — Mission Status	https://science.nasa.gov/mission/dawn/mission-status/
NASA Science mission page — Overview	https://science.nasa.gov/mission/dawn/science/overview/
NASA Science mission page — Spacecraft	https://science.nasa.gov/mission/dawn/technology/spacecraft/
NASA Science mission page — End Of Mission	https://science.nasa.gov/mission/dawn/toolkit/end-of-mission/
NASA Science mission page — Insight	https://science.nasa.gov/mission/insight/
NASA Science mission page — Science Instruments	https://science.nasa.gov/mission/insight/science-instruments/
NASA Science mission page — Kepler	https://science.nasa.gov/mission/kepler/
NASA Science mission page — In Depth	https://science.nasa.gov/mission/kepler/in-depth/
NASA Science mission page — Mars 2020 Perseverance	https://science.nasa.gov/mission/mars-2020-perseverance/
NASA Science mission page — Ingenuity Mars Helicopter	https://science.nasa.gov/mission/mars-2020-perseverance/ingenuity-mars-helicopte
NASA Science mission page — Mars Exploration Rovers Spirit And Opportunity	https://science.nasa.gov/mission/mars-exploration-rovers-spirit-and-opportunity/
NASA Science mission page — Science Instruments	https://science.nasa.gov/mission/mars-exploration-rovers-spirit-and-opportunity/science-instruments/
NASA Science mission page — Mars Pathfinder	https://science.nasa.gov/mission/mars-pathfinder/
NASA Science mission page — Mars Phoenix	https://science.nasa.gov/mission/mars-phoenix/
NASA Science mission page — Mer Opportunity	https://science.nasa.gov/mission/mer-opportunity/
NASA Science mission page — Mer Spirit	https://science.nasa.gov/mission/mer-spirit/
NASA Science mission page — Near Shoemaker	https://science.nasa.gov/mission/near-shoemaker/
NASA Science mission page — Pioneer 10	https://science.nasa.gov/mission/pioneer-10/
NASA Science mission page — Pioneer 11	https://science.nasa.gov/mission/pioneer-11/
NASA Science mission page — Surveyor 3	https://science.nasa.gov/mission/surveyor-3/
NASA Science mission page — Viking 1	https://science.nasa.gov/mission/viking-1/
NASA Science mission page — Viking 2	https://science.nasa.gov/mission/viking-2/
NASA Science mission page — Viking	https://science.nasa.gov/mission/viking/
NASA Science mission page — Spacecraft And Science	https://science.nasa.gov/mission/viking/spacecraft-and-science/
NASA mission/article page — Pioneer 10 Sends Last Signal	https://science.nasa.gov/missions/pioneer/pioneer-10-sends-last-signal/

Link Text	Link Url
NASA Science resource — Kepler Space Telescope	https://science.nasa.gov/resource/kepler-space-telescope/
NASA Science resource — Sojourner Rover Near The Dice	https://science.nasa.gov/resource/sojourner-rover-near-the-dice/
NASA Science solar-system article — Jpl Navigators Guide Near To Historic Landing On Asteroid Eros	https://science.nasa.gov/solar-system/jpl-navigators-guide-near-to-historic-landing-c asteroid-eros/
NASA Science solar-system article — Near Shoemakers Historic Landing On Eros Exceeds Science Engineering Expectations	https://science.nasa.gov/solar-system/near-shoemakers-historic-landing-on-eros-exceeds-science-engineering-expectations/
NASA Science universe article — Goodnight Kepler Final Commands For Space Telescope	https://science.nasa.gov/universe/exoplanets/goodnight-kepler-final-commands-for-space-telescope/
www-robotics.jpl.nasa.gov/what-we-do/flight-projects/pathfinder/	https://www-robotics.jpl.nasa.gov/what-we-do/flight-projects/pathfinder/
www.giss.nasa.gov/tools/mars24/help/landers.html	https://www.giss.nasa.gov/tools/mars24/help/landers.html
NASA JPL mission page — Ingenuity	https://www.jpl.nasa.gov/missions/ingenuity/
NASA JPL mission page — Mars Exploration Rover Opportunity Mer	https://www.jpl.nasa.gov/missions/mars-exploration-rover-opportunity-mer/
NASA JPL mission page — Mars Pathfinder Sojourner Rover	https://www.jpl.nasa.gov/missions/mars-pathfinder-sojourner-rover/
NASA JPL mission page — Phoenix	https://www.jpl.nasa.gov/missions/phoenix/
NASA JPL mission page — Surveyor 3	https://www.jpl.nasa.gov/missions/surveyor-3/
NASA JPL news — Nasa Retires Insight Mars Lander Mission After Years Of Science	https://www.jpl.nasa.gov/news/nasa-retires-insight-mars-lander-mission-after-years-science/
NASA JPL news — Nasas Opportunity Rover Mission On Mars Comes To End	https://www.jpl.nasa.gov/news/nasas-opportunity-rover-mission-on-mars-comes-to-e
NASA gallery — Apollo 15	https://www.nasa.gov/gallery/apollo-15/
NASA history article — 10 Years Ago Kepler Planet Finder Is Launched	https://www.nasa.gov/history/10-years-ago-kepler-planet-finder-is-launched/
NASA history article — 50 Years Ago First Apollo Lunar Surface Experiment Package Accepted	https://www.nasa.gov/history/50-years-ago-first-apollo-lunar-surface-experiment-package-accepted/
NASA mission/article page — Apollo 11 Mission Overview	https://www.nasa.gov/missions/apollo/apollo-11/apollo-11-mission-overview/
NASA mission/article page — Apollo 15 Mission Details	https://www.nasa.gov/missions/apollo/apollo-15-mission-details/
NASA mission/article page — Lunar Reconnaissance Orbiter Looks At Apollo 12 Surveyor 3 Landing Sites	https://www.nasa.gov/missions/lunar-reconnaissance-orbiter-looks-at-apollo-12-surveyor-3-landing-sites/
NASA news release — After Three Years On Mars Nasas Ingenuity Helicopter Mission Ends	https://www.nasa.gov/news-release/after-three-years-on-mars-nasas-ingenuity-helicopter-mission-ends/
NASA solar-system article — 40 Years Ago Pioneer 11 First To Explore Saturn	https://www.nasa.gov/solar-system/planets/saturn/40-years-ago-pioneer-11-first-to-explore-saturn/

6. Space Agency Partner & Other Data

Space-agency partner: None claimed. The NASA links are listed in Section 7. The following are the non-NASA tools, platforms, libraries, fonts, and other data sources actually used by the project.

Link Text	Link Url
Live project	https://lunar-relics-3d.vercel.app/
Public source repository	https://github.com/yousofahmad/lunar-relics-3d
30-second demo video	https://lunar-relics-3d.vercel.app/demo/lunar-relics-3d-demo-30s.mp4
NASA Space Apps Challenge 2026 challenge page	https://www.spaceappschallenge.org/2026/challenges/nasas-discarded-equipment-storytelling/
Manus SI agent and Cue companion tool	https://manus.im
React	https://react.dev/
React DOM	https://www.npmjs.com/package/react-dom
Vite	https://vite.dev/
Three.js	https://threejs.org/
@react-three/fiber	https://github.com/pmndrs/react-three-fiber
@react-three/drei	https://github.com/pmndrs/drei
Tailwind CSS	https://tailwindcss.com/
PostCSS	https://postcss.org/
Autoprefixer	https://github.com/postcss/autoprefixer
lucide-react	https://lucide.dev/guide/packages/lucide-react
canvas-confetti	https://www.npmjs.com/package/canvas-confetti
Web Audio API	https://developer.mozilla.org/en-US/docs/Web/API/Web_Audio_API
Vercel hosting	https://vercel.com/
GitHub	https://github.com/
FFmpeg	https://ffmpeg.org/
Python	https://www.python.org/
Playwright	https://playwright.dev/
Node.js test runner	https://nodejs.org/api/test.html

Link Text	Link Url
Cairo font	https://fonts.google.com/specimen/Cairo
DM Mono font	https://fonts.google.com/specimen/DM+Mono
Inter font	https://fonts.google.com/specimen/Inter
Orbitron font	https://fonts.google.com/specimen/Orbitron
Google Fonts stylesheet used by the site	https://fonts.googleapis.com/css2?family=Cairo:wght@400;500;600;700&family=DM+Mono:wght@400;500&family=Inter:wght@400;500;600;700&family=Orl
USGS Mars Viking colorized global mosaic	https://astrogeology.usgs.gov/search/map/mars_viking_colorized_global_mosaic_232m
USGS Mars mosaic dataset	https://astrogeology.usgs.gov/ckan/dataset/7131d503-cdc9-45a5-8f83-5126c0fd397e/resource/6afad901-1caa-48a7-8b62-3911da0004c2/download/mars_viking_mdim21_clrmosaic_global_1024.jpg

7. Use of Super Intelligence (SI)

The project idea and concept—spacecraft telling their own stories in Arabic and English as a 3D archive—came from our team, and the first version of the site existed before SI was used. SI then helped build and refine it: research and analysis, mission-data writing, Arabic–English translation, React/Three.js code, asset optimization, narration audio, QA, and demo-video preparation—with the SI agent Manus and its companion tool Cue, directed and reviewed by the team. Five spacecraft 3D models (Apollo 15 rover, ALSEP, Surveyor 3, Sojourner, Phoenix) were generated with SI from NASA's published specifications and are labeled in the app as "SI-generated · not from NASA"; the other eleven models are unmodified NASA-published meshes. The 30-second demo video carries a visible "SI-generated" watermark and SI metadata. Narration MP3s acknowledge SI generation in ID3 metadata, and the site labels the narration as an AI-generated voice. The public repository includes an SI disclosure section in its README, and source files carry SI notes. No NASA logos, flags, mission identifiers, or insignia were generated or modified. The team reviewed the output for provenance, language, accessibility, and factual limits before submission.